

Project Design Phase-II

Data Flow Diagram & User Stories

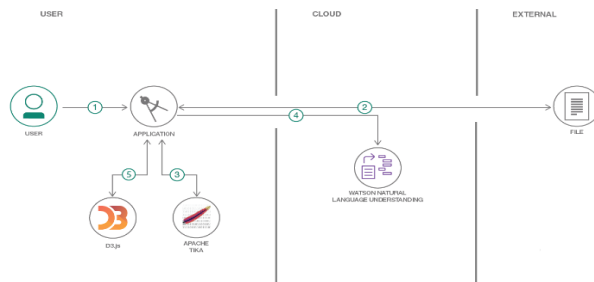
Date	5 September 2026
Team ID	SWUID2026190414
Project Name	Metro Ticket Generating System in ServiceNow
Maximum Marks	4 Marks

Data Flow Diagrams:

A Data Flow Diagram (DFD) is a traditional visual representation of the information flows within a system. A neat and clear DFD can depict the right amount of the system requirement graphically. It shows how data enters and leaves the system, what changes the information, and where data is stored.

DFD Level 0 – Metro Ticket Generating System

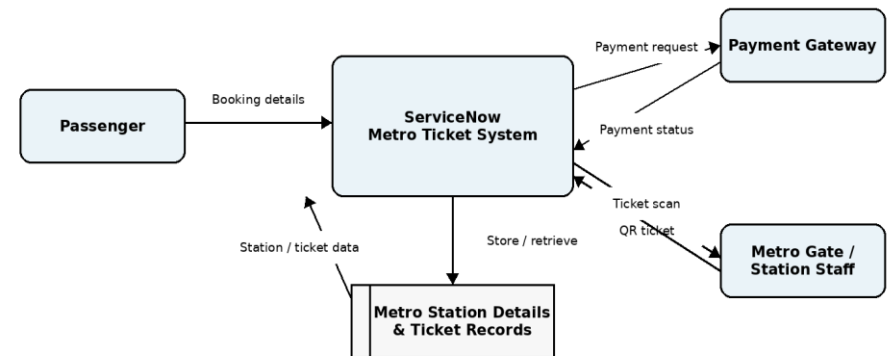
Flow



1. User configures credentials for the Watson Natural Language Understanding service and starts the app.
2. User selects data file to process and load.
3. Apache Tika extracts text from the data file.
4. Extracted text is passed to Watson NLU for enrichment.
5. Enriched data is visualized in the UI using the D3.js library.

Example: DFD Level 0 (Industry Standard)

DFD Level 0 - Metro Ticket Generating System



1. Passenger selects source, destination, journey type and number of passengers.
2. System validates the request and calculates the fare.
3. Payment is processed through the selected payment mode.
4. A digital QR ticket is generated and shown to the passenger.
5. Ticket and station information are maintained in ServiceNow records.

User Stories

Use the below template to list all the user stories for the product.

User Type	Functional Requirement (Epic)	User Story Number	User Story / Task	Acceptance criteria	Priority	Release
Passenger	Ticket Booking	USN-1	As a passenger, I can select my starting station and destination station to book a metro ticket.	I can select valid source and destination stations.	High	Sprint-1
	Journey Selection	USN-2	As a passenger, I can choose Single Journey or Return Journey.	The selected journey type is stored correctly.	High	Sprint-1
	Passenger Details	USN-3	As a passenger, I can select the number of passengers for the ticket.	I can select the required number of passengers and continue booking.	High	Sprint-1
	Fare Calculation	USN-4	As a passenger, I can see the fare calculated automatically for my selected journey and passengers.	The system calculates and displays the correct payable amount.	High	Sprint-1
	Payment	USN-5	As a passenger, I can select a payment mode such as UPI, Card, or Others.	The selected payment mode is captured with the ticket request.	High	Sprint-2
	Ticket Generation	USN-6	As a passenger, I can submit my booking request and receive a digital metro ticket.	A ticket request is created successfully after submission.	High	Sprint-2
	QR Code	USN-7	As a passenger, I can view a QR code for my metro ticket.	A QR code is displayed for the generated ticket and can be scanned at the gate.	High	Sprint-2
Metro Station Staff	Station Details	USN-8	As station staff, I can maintain metro station details used for ticket booking.	Station records are available and can be maintained in ServiceNow.	Medium	Sprint-2
Administrator	Access Control	USN-9	As an administrator, I can manage access to metro station and ticket-related records.	Only authorized users can perform permitted record operations.	Medium	Sprint-3
Administrator	Deployment	USN-10	As an administrator, I can package the completed project configuration in an Update Set.	The Update Set contains the required project configuration for deployment.	Medium	Sprint-3